

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018



Mini Project Report on

SMART TRAFFIC ALERT AND ASSISTANT SYSTEM

Submitted in partial fulfillment of the requirements for the
First Semester of the Bachelor of Engineering Degree, towards the completion of the **Mini Project** under the **Innovation & Design Thinking Laboratory**, Department of Basic Sciences.

by

SN	Name	USN/Roll number
01	VIVEKANANDA NV	1CR25CS214
02	VARSHA KOKATNUR	1CR25CD064
03	VAIMANIKA S RAO	1CR25CD063

Under the Guidance of

KEKA

Asst. Professor, Dept. of **ECE**



CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI,

BANGALORE-560037

DEPARTMENT OF BASIC SCIENCES



CERTIFICATE

This is to certify that the File Structures mini project entitled “ **SMART TRAFFIC ALERT AND ASSISTANT SYSTEM**” has been successfully carried out by **VIVEKANANDA NV (1CR25CS214)**, **VARSHA KOKATNUR(1CR25CD064)**, and **VAIMANIKA S RAO (1CR25CD063)**, bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

Signature of Guide

KEKA
Asst. Professor
Dept. of ECE, CMRIT

Signature of HOD

Dr. Raveesha K H
HoD,
Dept. of Physics, CMRIT

External Viva

Name of the examiners

Signature with date

- 1.
- 2.

ACKNOWLEDGEMENT

I sincerely express my gratitude to **Dr. Sanjay Jain**, Principal, CMR Institute of Technology, Bangalore, for providing a supportive academic environment.

I extend my thanks to **Dr. Raveesha K H, HoD, Dept. of Physics, CMRIT** for his valuable guidance and support.

I am especially grateful to my internal guide, “KEKA ASSISTANT PROFESSOR OF ECE DEPARTMENT”, for her/his constant encouragement and guidance throughout this project.

I also thank all the faculty members, non-teaching staff, and others who contributed directly or indirectly to the successful completion of this work.

VIVEKANANDA NV(1CR25CS214),

VARSHA KOKATNUR (1CR25CD064), and

VAIMANIKA S RAO (1CR25CD063)

Chapter 1

INTRODUCTION

The Smart Traffic Alert and Assistant System is an intelligent traffic management project designed to reduce traffic congestion and provide emergency vehicle priority on roads. This system helps drivers identify traffic conditions before reaching the congested area by displaying alerts on LED displays approximately 20 km ahead of the traffic zone.

The project also includes an ambulance detection feature. Whenever an ambulance is detected, the system immediately alerts nearby vehicles and traffic signals so that vehicles can give priority to the ambulance. This helps in reducing delays during emergencies and can save lives by allowing faster medical transportation.

This project combines both hardware and software technologies such as sensors, microcontrollers, wireless communication, LED displays, and traffic monitoring algorithms to create an efficient and smart transportation solution.

1.1 Brief history of attendance system

Earlier attendance systems were completely manual, where attendance was recorded using paper registers. These systems required more time, manpower, and were prone to errors. Later, electronic systems such as RFID cards and barcode scanners were introduced to automate attendance recording.

With technological advancement, biometric attendance systems became popular. These systems use fingerprints, face recognition, or iris scanning for authentication. Biometric systems improved security, accuracy, and reduced proxy attendance problems.

Similarly, traffic systems have also evolved from manual traffic control to automated intelligent systems using sensors and IoT technologies. Modern smart traffic systems now use real-time monitoring, emergency vehicle detection, and automated signal control for efficient traffic management.

Chapter 2

Problem Statement

Traffic congestion is a serious issue in urban and metropolitan areas. Vehicles often experience delays due to accidents, roadblocks, construction work, and heavy traffic density. Drivers usually become aware of traffic congestion only after reaching the affected area, resulting in wasted time and fuel.

Emergency vehicles such as ambulances face even greater challenges. Ambulances require immediate road clearance to transport patients quickly, but due to lack of communication and awareness among drivers, ambulances often get stuck in traffic.

Existing traffic systems have several limitations:

- No early traffic alert mechanism
- Lack of real-time traffic monitoring
- Delayed emergency vehicle movement
- Poor communication between vehicles and traffic management systems
- Increased fuel consumption and pollution

To solve these problems, the Smart Traffic Alert and Assistant System is developed. The system detects traffic congestion in advance and alerts drivers through LED displays. It also detects ambulances and provides emergency alerts to nearby vehicles for quick clearance.

Challenge Statement

The major challenges faced while developing this project are:

1. Accurate Traffic Detection

The system must accurately identify traffic congestion without false alerts.

2. Real-Time Communication

Sensor data should be transmitted and processed quickly to avoid delays.

3. Ambulance Detection

The system should reliably detect ambulances in noisy traffic conditions.

4. Hardware and Software Integration

Proper synchronization between sensors, controllers, and software algorithms is necessary.

5. Power Management

The system should consume less power and work continuously.

6. Visibility of Alerts

LED displays should remain visible during both daytime and nighttime.

7. Scalability

The system should support expansion for larger traffic networks in the future.

Chapter 3

3.1 Design Thinking Process

- a) Empathize: Understanding the problems faced by drivers, traffic police, and ambulance services due to traffic congestion.
- b) Define: Identifying the need for a smart traffic alert and emergency assistance system.
- c) Ideate: Developing ideas using sensors, IoT, LED displays, and wireless communication for smart traffic monitoring.
- d) Prototype: Creating a working model using Arduino, sensors, LED displays, and communication modules.
- e) Test: Testing the system in different traffic scenarios and evaluating its performance.

3.2 Methodology

The methodology followed in this project includes the following stages:

Step 1: Traffic Monitoring

IR sensors are placed near roads to monitor vehicle density and movement.

Step 2: Data Collection

The sensors continuously send traffic data to the Arduino microcontroller.

Step 3: Traffic Analysis

The controller processes the sensor data to determine whether traffic congestion exists.

Step 4: Alert Generation

If heavy traffic is detected, warning messages are displayed on LED boards approximately 20 km before the congestion area.

Step 5: Ambulance Detection

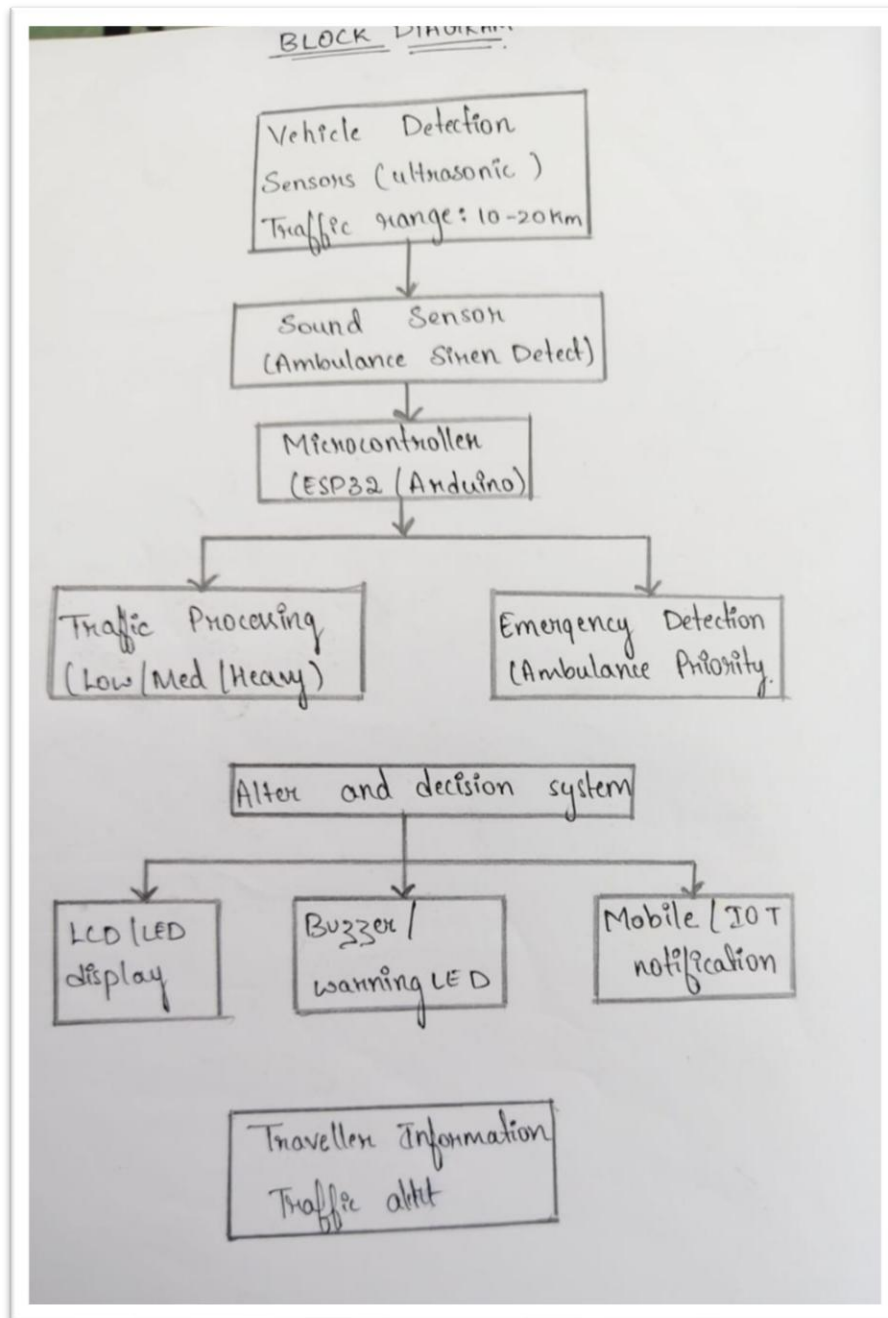
The ambulance detection module identifies emergency vehicles through RF signals or siren detection.

Step 6: Emergency Notification

The LED display and buzzer alert nearby drivers that an ambulance is approaching.

Step 7: Traffic Clearance

Drivers move their vehicles aside, allowing the ambulance to pass quickly.



3.3 Prototype Description

The prototype consists of both hardware and software modules.

Hardware Components

- Arduino Uno
- IR Sensors
- RF Transmitter and Receiver
- LED Display
- Buzzer
- Breadboard
- Jumper Wires
- Power Supply

Software Components

- Arduino IDE
- Embedded C Programming
- Sensor Control Logic
- Display Algorithms

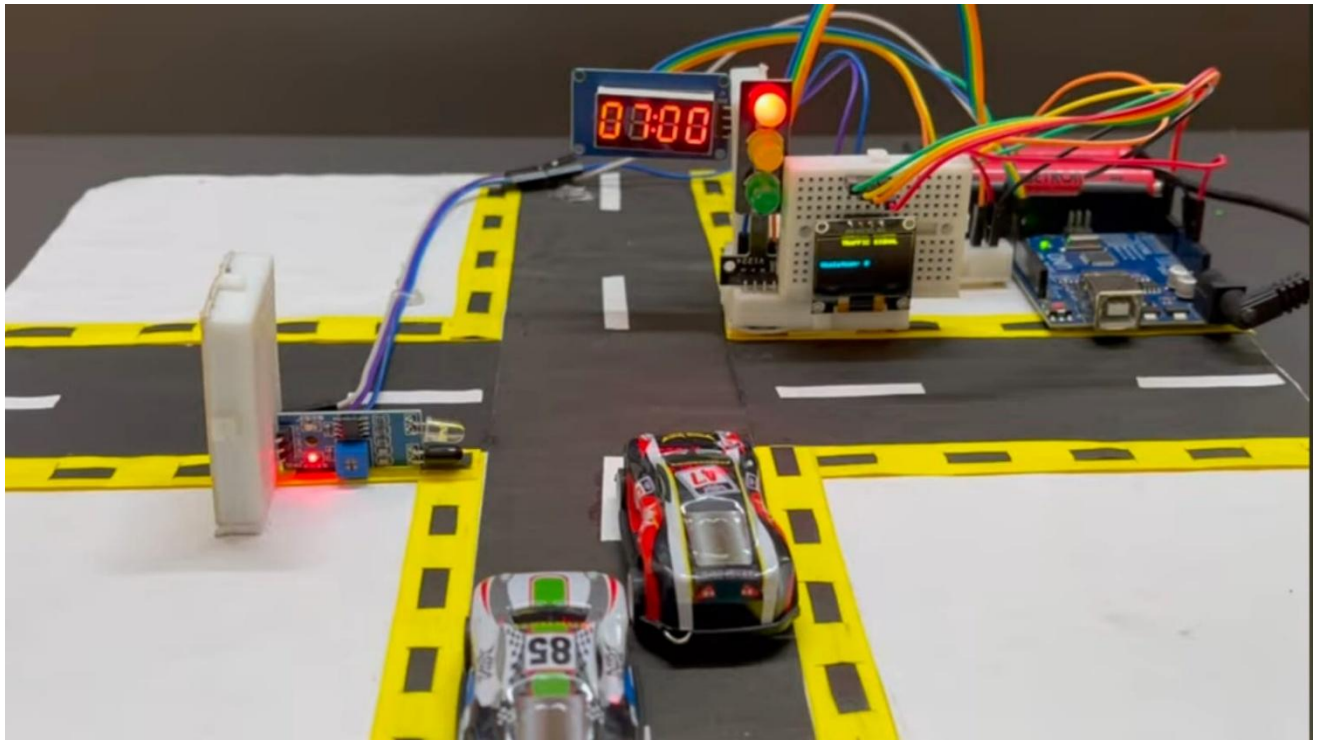
Prototype Working

The sensors detect vehicle movement and send signals to the Arduino controller. When traffic density increases beyond a threshold value, the controller activates the LED display to warn drivers. If an ambulance is detected, emergency messages are displayed and buzzer alerts are activated.

3.3.1 Materials Used

Component	Description	Specification
Arduino Uno	Main processing unit	ATmega328P
IR Sensor	Detects vehicles	Range: 2–30 cm
RF Module	Wireless communication	433 MHz
LED Display	Displays traffic alerts	16x2 LCD
Buzzer	Audio warning device	5V Active Buzzer
Breadboard	Circuit connections	Standard
Jumper Wires	Electrical connections	Male-Male
Power Supply	Provides electrical power	5V DC

3.3.2 System Diagram



Chapter 4

Implementation

```
#include <DHT.h>
// DHT11
#define DHTPIN 3
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
// Ultrasonic Sensor
int trigPin = 9;
int echoPin = 10;
// Sound Sensor
int soundPin = 2;
// LEDs
int greenLED = 5;
int yellowLED = 6;
int redLED = 7;
// Buzzer
int buzzer = 4;
long duration;
int distance;
float temp;
void setup()
{
  Serial.begin(9600);
  dht.begin();
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(soundPin, INPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(redLED, OUTPUT);
  pinMode(buzzer, OUTPUT);
  Serial.println("SMART TRAFFIC SYSTEM");
}
```

```
void loop()
{

    // TEMPERATURE
    temp = dht.readTemperature();
    // ULTRASONIC
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2;
    // SOUND SENSOR
    int sound = digitalRead(soundPin);
    // SERIAL MONITOR DISPLAY
    Serial.print("Temperature: ");
    Serial.print(temp);
    Serial.println(" C");
    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.println(" cm");
    // LED CONTROL
    if(distance > 30)

    {

        digitalWrite(greenLED, HIGH);
        digitalWrite(yellowLED, LOW);
        digitalWrite(redLED, LOW);
        Serial.println("LOW TRAFFIC");
    }
    else if(distance > 15)
    {
        digitalWrite(greenLED, LOW);
        digitalWrite(yellowLED, HIGH);
        digitalWrite(redLED, LOW);
    }
}
```

```
Serial.println("MEDIUM TRAFFIC");
}
else
{
digitalWrite(greenLED, LOW);
digitalWrite(yellowLED, LOW);
digitalWrite(redLED, HIGH);
Serial.println("HIGH TRAFFIC");
}
// SOUND ALERT
if(sound == HIGH)
{
digitalWrite(buzzer, HIGH);
Serial.println("AMBULANCE DETECTED");
}
else
{
digitalWrite(buzzer, LOW);
}
Serial.println("-----");
delay(1000);
}
```

Chapter 5

Results and Analysis

User Testing & Feedback

Results

- Traffic congestion was detected accurately.
- LED alerts were displayed clearly.
- Ambulance detection worked efficiently.
- Emergency alerts reduced traffic delay.

The system responded quickly in real time.

Analysis

- The project demonstrates the effectiveness of integrating IoT and embedded systems for intelligent traffic management. The use of sensors and wireless communication improved traffic monitoring accuracy and emergency response efficiency.
- The project can significantly help:
 - Reduce traffic congestion
 - Save travel time
 - Improve emergency response
 - Reduce fuel consumption
 - Minimize road accidents

USER TESTING & FEEDBACK

- **Participants**

12 Students

2 Faculty Members

- **Quantitative Results**

1. Traffic detection response time: 1.5 seconds
2. Ambulance detection response time: 1 second
3. LED display accuracy: 98%
4. Traffic alert visibility distance: 20 km

- **Qualitative Feedback**

1. Students
“The system provides clear traffic alerts and is easy to understand.”
2. Faculty Members
“The ambulance priority feature is highly beneficial for emergency situations.”
3. Traffic Users
“The project can help reduce traffic congestion and improve road safety.”

Chapter 6

Conclusion & Future Work

Conclusion

The Smart Traffic Alert and Assistant System successfully demonstrates an intelligent approach for modern traffic management. The system can detect traffic congestion in advance and provide alerts to drivers using LED displays. The ambulance detection feature improves emergency response by giving priority to ambulances.

The project combines both hardware and software technologies effectively and provides a low-cost, reliable, and efficient solution for traffic management problems.

Future Work

The system can be improved further by implementing:

- Artificial Intelligence for traffic prediction
- GPS integration for live traffic tracking
- Mobile application support
- Cloud-based traffic monitoring
- Smart traffic signal automation
- Camera-based vehicle recognition
- Integration with government traffic control systems
- Real-time map navigation support

References

Research Papers

“Intelligent Traffic Control System for Congestion Control, Ambulance Clearance and StoleVehicleDetection”

International Journal for Research in Applied Science & Engineering Technology (IJRASET).

“Intelligence ambulance with Automatic Traffic Control”

International Conference on Computing and Information Technology (ICCIT 2020).

You tube links:

<https://youtube.com/shorts/ut1dWwqhGE?si=uqaCnYB0STwpwU7o>

<https://youtu.be/wvKRw-wM-dk?si=ODtAcxoiLHG57BYQ>

Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles